A Predictive Model for Assessing Access to Depression Treatment, Medication Compliance, and Recovery Rates

Group 29

Introduction

The objective of our project is to create a predictive model that demonstrates the relationship between the prevalence of depression in England, the treatment received by patients, and the rate of antidepressant-talk therapy and recovery. This model aims to forecast the pressures on the British health system and the demand for interactive therapy

Background

With the advancement of technology and increased social competition, people are facing greater pressures at work and in life. Serious mental illness has become a major concern in British society. Due to social and economic factors, 56% of employees in the British workforce experience symptoms of depression, with 25% of these symptoms reaching clinically relevant levels (Champion Health, 2023). In 2024, the Royal College of Psychiatrists reported that people with severe mental illness live 15 to 20 years less than the general population. Severe mental illness affects not only personal health but also social stability and economic development in England. As a result, various antidepressant medications and treatment programs have been developed and implemented to address this problem.

On the other hand, the growing demand for healthcare poses a great challenge to the existing healthcare system. According to the National Health Service in 2022, the use of antidepressant treatment has shown a year-on-year increase in England. In 2022, an estimated 83.4 million antidepressant prescriptions were issued, marking an 8.4% increase from 2021. However, only around 10% of people with depression sought support. The unmet need for mental health treatment worsens the problem. Serious mental illness poses a critical challenge to the British government, the public, and the health system. It increases social service costs, reduces public well-being, and intensifies conflicts over healthcare resources. Therefore, planning and predicting the prevalence and recovery rates of serious mental illness are crucial. These efforts help rationalize the allocation of public resources and improve the physical and mental health of the population.

NHS Talking Therapies is a free service provided by NHS England to deliver short-term therapeutic support for adults suffering from common mental health problems such as depression and anxiety. The service is largely based on Cognitive Behavioral Therapy (CBT) and helps patients learn skills to manage anxiety or low mood, improving their mental health. Patients can access the service through GP referral or self-referral in the form of face-to-face, telephone or online counseling. A systematic evaluation published in *World Psychiatry* has shown that online cognitive behavioral therapy achieves similar clinical outcomes and is more cost-effective than traditional face-to-face therapy. In addition, personalized self-referral chatbots have been introduced into NHS Talking Therapies, significantly improving access to mental health services, particularly for long-neglected groups.

Approach

To predict the prevalence and recovery of the population with depression in England, along with their treatment utilization, we will develop a dynamic systems model based on a modified version of the Leslie Matrix. The Leslie Matrix, proposed by Patrick H. Leslie in 1945, is a mathematical tool used for modeling population dynamics. The Leslie Matrix is particularly well suited for describing the dynamics of stage-specific human or biological populations, dynamically predicting population structures and distributions over time, and analyzing inflows and outflows between various categories. The core of the Leslie Matrix is based on population dynamics at different stages (age groups or states). In the case of depression, although it does not involve an age-based fertility and survival relationship, state transitions, such as prevalence, treatment, and recovery, can be represented by a similar dynamic matrix to inform our research.

Our model demonstrates the process through which the population with depression receives treatment, engages with NHS Talking Therapies, and follows possible paths to eventual cure or relapse. The model aims to predict key demographic trends over the next 5 years. Historical data from official databases and literature will be used to calibrate and optimize the model, serving as initial conditions. The model will also help assess long-term trends in population behavior and provide insights into its potential impact on government policy-making and healthcare resource allocation.

Hypothesis

Our hypotheses are as follows:

Driven by underlying factors, the total population with depression is expected to grow. Consequently, the demand for treatment, including NHS Talking Therapies, will also rise. With growing social attention, accelerated medication development, and improved healthcare, the recovering population is expected to grow the fastest. This will also result in increased demand for post-recovery psychological counseling and ongoing monitoring of depression relapse.

Data

Given the complexity of the topic and many prejudiced actors it was critical to choose the right source. For our project the most authoritative and reliable source is National Health Service (data accessed through NHS digital portal). NHS was chosen as a primary source of raw data, that has not undergone interpretation. The data is provided as is. For other aspects of the project (e.g. construction of the Leslie matrix and optimization modeling) other available scientific sources were used.

Our dataset focuses on the time period starting from 2017 up to 2023. The figures for the "New Referrals" category are collected from monthly registration and added up to form a yearly period. An important notification should be made regarding this category: these cases might include when a single person was referred to several services. Thus, this data does not necessarily imply that each number represents a unique patient. Patients were chosen into the "Recovered" category of our model only if they belonged to the category defined by the NHS as "reliable recovery". Addition to this category the patients who just showed certain aspects of improvement would broaden the category excessively and cause erroneous conclusions and misinterpretation of model results. The same logic was applied for the "Relapse" category of the model. Only patients who revealed reliable deterioration rates were included into this category.

The table with relevant raw data is provided in the appendix 1.

Methodology

The aim of this study was to predict the morbidity, treatment, medication and recovery of patients with depression in England by constructing a modified dynamic model based on the Leslie Matrix.

Given the fact that depression treatment is a lengthy process with certain challenges to assess metrics, such as recovery or relapse, we rely on the assessment suggested by NHS Talking Therapy (previously known as Improving Access to Psychological Therapies - IAPT). Emotional recovery is quite subjective and difficult to quantify. Thus, based on the therapy standards, if patient shows steady improvement on Patient Health Questionnaire-9 (PHQ-9) and General Anxiety Disorder-7 (GAD-7) questionnaires, only then he/she can fall into the "recovered" category, while steady deterioration of the results place such a patient into "relapsed" category, as patients finish the therapy and start the cycle again by new referrals.

Model

Mental health conditions, such as depression, involve complex pathways influenced by various factors, including symptoms severity and treatment options. In this study, a simplified population model was developed to illustrate the key transitions between different states, such as experiencing symptoms, receiving treatment, getting NHS talking therapy, achieving recovery, and experiencing mental illness relapse.

The data for the model was mainly sourced from NHS England Mental Health Data Hub, which offers reliable and representative information on patient pathways and outcomes. To validate this approach, this model prediction, as shown in Fig. 1, was compared with the historical data, confirming the accuracy of the assumptions made.

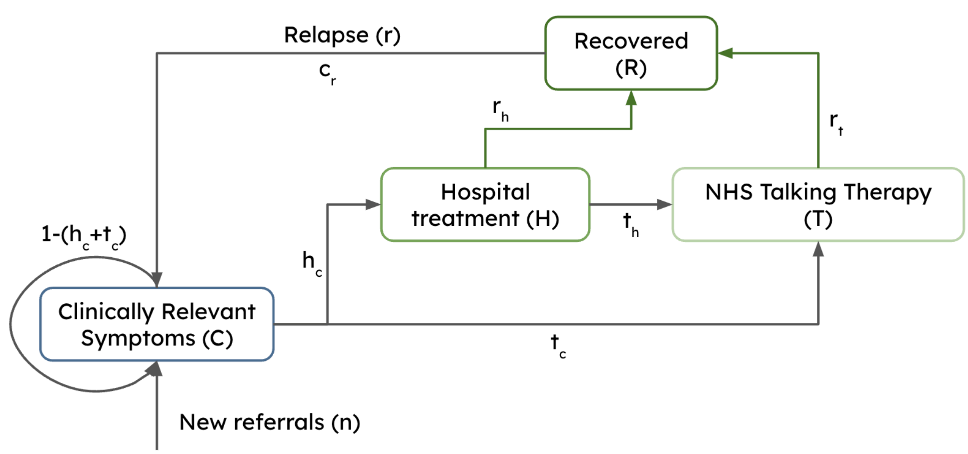

Fig. 1 Clinical pathway diagram for depression treatment and recovery

From Fig. 1, the model consists of four main states:

- Clinically relevant symptoms (C): Individuals experiencing significant depressive symptoms but have not yet started proper treatment.
- Hospital treatment (H): Individuals undergoing treatment for mental illness.
- NHS talking therapy (T): Individuals receiving talking therapy, this service can be accessed both through direct self-referral or through a referral from GP.
- Recovered (R): Individuals who have recovered from depressive symptoms or mental illness and no longer need any active treatment.

Assumptions

<u>Grouping of Depression Subtypes</u>

According to Benazzi (2016), accurately subtyping of depression is crucial for ensuring appropriate and effective treatment. Examples of depression subtypes include bipolar II depression, seasonal depression, and mixed depression. All forms of depression subtypes are treated as a single group in the model, as the focus is on trends within the overall population rather than subtype-specific outcomes.

<u>Referral Consistency</u>

For the input variable of new referrals, the trend was estimated based on raw data of new referrals, as shown in Appendix A. The model assumes that the rate of new referrals is consistent across time periods, even though, in reality, the rate may vary due to various factors such as increase in public awareness or change in healthcare-related policy.

<u>Uniform treatment effectiveness</u>

In this model, treatment effectiveness is assumed to be uniform across all individuals, regardless of the factors such as treatment and medication plans.

<u>Linear rates between states</u>

Transition rates between each state are assumed to be linear over time. However, in real-world scenarios, these rates may follow non-linear patterns influenced by various external factors, such as treatment effectiveness.

Relapse rate estimation

For the relapse rate (c_r), the model uses data from research on depression recurrence. However, it may account for the variations based on depression subtypes, treatment plans, or individual differences.

Leslie Matrix

A Leslie Matrix, M, was constructed to represent the relationship between the states of mental health treatment and recovery model.

$$M = \begin{bmatrix} 1-(h_c+t_c) & 0 & 0 & c_r \\ h_c & 1-(t_h+r_h) & 0 & 0 \\ t_c & t_h & 1-r_t & 0 \\ 0 & r_h & r_t & 0 \end{bmatrix}$$

According to Sverdlove (1981), the equation $V_{n+t} = M^t V_n + Y^t$ was used to calculate the population in each state at the next time step, where V_n is the initial state vector, t is the number of years, and Y is nonnegative input vector representing new referrals.

$$\begin{bmatrix} 1-(h_c+t_c) & 0 & 0 & c_r \\ h_c & 1-(t_h+r_h) & 0 & 0 \\ t_c & t_h & 1-r_t & 0 \\ 0 & r_h & r_t & 0 \end{bmatrix}^t \begin{bmatrix} C_n \\ H_n \\ T_n \\ R_n \end{bmatrix} + \begin{bmatrix} n \\ 0 \\ 0 \\ 0 \end{bmatrix}^t = \begin{bmatrix} C_{n+t} \\ H_{n+t} \\ T_{n+t} \\ R_{n+t} \end{bmatrix}$$

Variables were assigned to each element in the model for depression treatment and recovery model, using available data where possible and estimated data based on existing research findings. The model was validated using historical data spanning a six-year period (see Appendix 1), to ensure its accuracy and reliability to further predict trends and dynamics in depression. Detailed information on the variables and method used to derive them is provided in Appendix 2.

$$M = \begin{bmatrix} 0.4913 & 0 & 0 & 0.0577 \\ 0.3597 & 0.1197 & 0 & 0 \\ 0.1490 & 0.6703 & 0.5199 & 0 \\ 0 & 0.2100 & 0.4801 & 0 \end{bmatrix} \quad Y^t = \begin{bmatrix} (1.08489^t)(n_n) \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Results

For model validation, raw data from 2020 was used to predict the outcomes for 2023. See Appendix 3 for the coding details. In conclusion, the model shows an increase in the total population with depression in England, the number of individuals with depression is projected to reach 7,410,858 by 2029, highlighting the need for demand for mental health services and effective treatment options.

The model also indicates an increase in the number of patients receiving hospital treatments and NHS talking therapies up to 2025, followed by a significant decline thereafter. This decline could suggest that, according to the model projection, there can possibly be barriers in accessing mental healthcare services or resources. However, this significant decline can also be from limitations in the model's assumptions related to treatment dynamics.

Overall, this model provides a structured approach in analysing depression treatment and recovery dynamics, with the assumptions on linear transition rates and uniform population behaviours which may not align with real-world situations. However, this model can be developed in the future to capture and present non-linear relationships of these elements to enhance its predictive accuracy and applicability.

Importance and Implications of the model

First, the predictive model established in this project provides strategic reference value for health policy makers and hospital managers in the UK. The model predicts the number of depression cases and treatment demand in the future, which enables the government and medical institutions to plan medical allocation in advance, such as expanding the number of trained psychotherapists, promoting self-screening programs or providing online consultation platforms. These interventions will reduce morbidity and avoid overloading the healthcare system.

Second, the prediction of the model can help decision-makers to methodically allocate medical resources and control costs. Through quantitative prediction of demand, they can judge the appropriate increase in talk therapy and other related investments, so as to make resource allocation more targeted, thus reducing the risk of resource waste and saving social medical costs.

Third, the model can be used to test "what-if scenarios". For newly psychological rehabilitation programs and innovative treatment models, the model can be used to estimate the impact of these measures on the overall treatment rate and recovery rate in the next few years, so as to provide data support and effectiveness estimation for policy reforms.

Finally, the high prevalence of depression affects not only the healthcare system, but also social stability and economic development. As an analytical tool, the model can produce indirect impact on other macro indicators, and can also be used to predict and analyze different diseases or other mental health problems, so as to provide a methodological blueprint for future research.

General Limitations

The topic of mental problems in general, and depression in particular, is a very difficult topic to study and build models on. For instance, during the process of data gathering we faced several limitations: there are both quantitative and qualitative metrics. While the case of new referrals seems to be obvious - each new referral is just added up, the case of “recovery” and “relapse” rates is more complex and multidimensional. The data there contains qualitative metrics and broken down into several categories, such as mild improvements, slight deterioration etc. One might also speculate regarding the improvement, questioning the sufficient time span to be considered healthy, if at all possible to recover fully from such conditions and not return back to them.

Moreover, due to the sensitivity of the topic and the nature of depression, the NHS does not strictly monitor each individual who refers to medical organizations with such problems. As a result, there are cases of double-entry (e.g. for new referrals), lack of sufficient tracking of patients (some patients drop out of the system after “hospital spell” stage), possibly lightweight approach to assign patients into “reliable recovery” category just after a several weeks of Talking therapy and undergoing a couple of psychological questionnaires. All these factors significantly contribute to data variation and projection inaccuracies.

Further Applications

The depression prediction model can be applied to different populations, such as groups defined by geographic region, gender, age, or socio-economic background. It can also be extended to other psychiatric disorders, such as anxiety or bipolar disorder, particularly those with similar dynamic characteristics.

In addition, the model can be used to analyze the effects of specific interventions, such as counseling services, improvements in medication, or the introduction of social support programs, to provide data-driven recommendations to policymakers and institutions. The model can also test the efficiency and effectiveness of other measures to cure depression by focusing on specific treatments. It can also help assess the long-term impact of healthcare resource allocation. For example, it can evaluate the benefits of providing more mental health services at the community level to improve overall mental health and reduce long-term healthcare costs.

Conclusion

This depression prediction model demonstrates that small changes in upstream factors (e.g., societal pressures, allocation of healthcare resources, development of antidepressant treatment programs) and initial conditions can lead to significant changes in depression prevalence, treatment, and recovery rates. These effects are especially pronounced in larger populations or when they have greater downstream impacts on healthcare resource demands.

For the healthcare system in England, this model indicates that depression treatment and rehabilitation services must be expanded to meet the anticipated demand. At the same time, priority should be given to developing long-term counseling and relapse monitoring programs. These initiatives aim to improve overall depression treatment outcomes and reduce pressure on the healthcare system.

Reference List:

Andersson, G. and Titov, n. (2014) 'Advantages and limitations of Internet-based interventions for common mental disorders', *World Psychiatry,* 13, pp. 4 - 14.

Benazzi, F. (2006) 'Various forms of depression', *Dialogues in Clinical Neuroscience*, 8(2), pp. 151 - 161.

Champion Health (2023) *Depression Statistics UK 2023 Data.* Available at: https://championhealth.co.uk/insights/depression-statistics/

Leslie, P. H., 1945. On the use of matrices in certain population mathematics. Biometrika, 33(3), pp.183–212.

National Institute for Health and Care Excellence (2024) *Depression: What is the prognosis?*. Available at: https://cks.nice.org.uk/topics/depression/background-information/prognosis/

NHS Bristol, North Somerset, and South Gloucestershire (no date) *Mental health services.* Available at: https://bnssg.icb.nhs.uk/health-and-care/mental-health-services/

NHS England (2024) *Mental Health Act statistics.* Available at: https://digital.nhs.uk/data-and-information/data-tools-and-services/data-services/mental-health-data-hub/dashboards/mental-health-act-statistics

Nuffieldtrust (2024) *NHS Talking Therapies (IAPT) programme.* Available at: https://www.nuffieldtrust.org.uk/resource/improving-access-to-psychological-therapies-iapt-programme#:~:text=In%202012%2F13%20there%20were,this%20fell%20to%201.7%20million

Public Health England (2018) *Severe mental illness (SMI) and physical health inequalities: briefing.* Available at: https://www.gov.uk/government/publications/severe-mental-illness-smi-physical-health-inequalities/severe-mental-illness-and-physical-health-inequalities-briefing#fn:6

Sverdlove, R. (1981) 'Leslie Models with Input', *Society for Industrial and Applied Mathematics*, 42(2), pp. 324 - 330.

Appendix 1

Raw data

Year	New referrals	Patients in contact with hospital services	Patients getting talking therapies			Reliable Recovery Rate	Relapse (Reliable Deteriorati on Rate)
			People getting talking therapies directly	Patients referred to talking therapies	Total		
2017	3,160,294	1,219,194	567,106	824,254	1,391,360	47%	N.A.
2018	3,494,448	1,316,146	554,709	885,248	1,439,957	48.3%	5.9%
2019	3,792,778	1,342,397	582,556	1,021,086	1,603,642	49.5%	5.8%
2020	3,825,428	1,371,842	606,192	1,088,598	1,694,790	48.5%	5.8%
2021	4,413,009	1,523,181	658,368	798,078	1,456,446	48.7%	5.4%
2022	4,644,210	1,660,613	664,087	1,148,509	1,812,596	47.3%	5.8%
2023	5,125,472	1,877,410	672,193	1,087,388	1,759,581	46.8%	5.9%

(https://www.nuffieldtrust.org.uk/resource/improving-access-to-psychological-therapies-iapt-programme#:~:text=In%202012%2F13%20there%20were,this%20fell%20to%201.7%20million.)
(https://digital.nhs.uk/data-and-information/publications/statistical/psychological-therapies-annual-reports-on-the-use-of-iapt-services)

Appendix 2

Increase rate of new referrals = 1.08489

Rate of patients getting hospital treatment(h_c) = 0.3597

Rate of patients getting talking therapy (t_c) = 0.1490

Rate of patients referred to talking therapy (t_h) = 0.6703

Average recovery rate of patients getting talking therapy (r_t) = 48.0143%

The recovery rate of normal depression patients in contact with hospitals (r_h) is approximately 21% (https://cks.nice.org.uk/topics/depression/background-information/prognosis/)

Average relapse rate (c_r) = 5.7667%

Coding part:

```
import pandas as pd
import numpy as np
#define new referrals and patients getting hospital treatments per year to the list
new_referrals = [3160294, 3494448, 3792778, 3825428, 4413009, 4644210, 5125472]
hospital_treatments = [1219194,1316146,1342397,1371842,1523181,1660613,1877410]

# Calculate a list of annual growth rates
growth_rates = [ ]
for i in range(1, len(new_referrals)):
  growth = hospital_treatments[i] / new_referrals[i]
  growth_rates.append(growth)

# Calculate the average growth rate
avg_growth_rate = sum(growth_rates) / len(growth_rates)
avg_growth_rate
print(avg_growth_rate)
```

0.3596998436821901

```
import pandas as pd
import numpy as np
#define new referrals and patients receiving talking therapies per year to the list
new_referrals = [3160294, 3494448, 3792778, 3825428, 4413009, 4644210, 5125472]
talking_therapies = [567106,554709,582556,606192,658368,664087,672193]

# Calculate a list of annual growth rates
growth_rates = [ ]
for i in range(1, len(new_referrals)):
   growth = talking_therapies[i] / new_referrals[i]
   growth_rates.append(growth)

# Calculate the average growth rate
avg_growth_rate = sum(growth_rates) / len(growth_rates)
avg_growth_rate
print(avg_growth_rate)
```

0.1490213409642714

```
import pandas as pd
import numpy as np
#define new referrals per year to the list
hospital_treatments = [1219194,1316146,1342397,1371842,1523181,1660613,1877410]
referred_to_talking_therapies = [824254,885248,1021086,1088598,798078,1148509,1087388]

# Calculate a list of annual growth rates
growth_rates = [ ]
for i in range(1, len(hospital_treatments)):
   growth = referred_to_talking_therapies[i] / hospital_treatments[i]
   growth_rates.append(growth)

# Calculate the average growth rate
```

```
avg_growth_rate = sum(growth_rates) / len(growth_rates)
avg_growth_rate
print(avg_growth_rate)
```

0.6702580542080706

```
#define recovery rates for each year
recovery_rate_therapy = [0.47, 0.483, 0.495, 0.485, 0.487, 0.473, 0.468]

#calculate the average recovery rate
rt = sum(recovery_rate_therapy) / len(recovery_rate_therapy)
print("Average recovery rate of patients getting talking therapy (rt):", rt)
```

0.4801428571428571

```
#define the annual recurrence rate
relapse_rates = [0.059, 0.058, 0.058, 0.054, 0.058, 0.059]  # Relapse rates from the table

#calculate the average recurrence rate was calculated
average_relapse_rate = sum(relapse_rates) / len(relapse_rates)
print("Average relapse rate (cr):", average_relapse_rate * 100, "%")
```

5.766666666666667 %

Appendix 3

```
#define the leslie matrix
M =
np.array([[0.4913,0,0,0.0577],[0.3597,0.1197,0,0],[0.1490,0.6703,0.52,0],[0,0.21,0.4801,0]])

#define initial state vector
V = np.array([5125472,1877410,1759581,1445725])

#define the input variable
Y = np.array([pow(1.08489,2)*4413009,0,0,0])

X = M@M
X@V+Y
```

array([6324179.54446917, 1017923.49777353, 2799932.44385265,

1546975.64498691])

	Predicted data for the year 2020	Actual data for the year 2020
C	6324179.54446917	5125472
H	1017923.49777353	1877410
T	2799932.44385265	1759581
R	1546975.64498691	1445725

Appendix 4

Year	People with depression (C)	Patients undergoing hospital treatment (H)	NHS Talking Therapy (T)	Recovered Patients (R)
2023	5125472	1877410	1759581	1445725
2024	7389192	2068358	2937105	1239031
2025	6543691	1183364	3301348	1844459
2026	6404478	627114	2711007	1833484
2027	6597176	351856	1944734	1433248
2028	6952701	216158	1319204	1007556
2029	7410858	141127	878618	678743